

Physics-Guided AI Digital Twin for Biogas Systems (GFIS)

Title Page

Physics-Guided AI Digital Twin for Biogas Systems (GFIS)

A Hybrid Artificial Intelligence Framework for Sustainable Biogas Optimization

Author: Akash S. Chatake

Program: M.Tech Artificial Intelligence & Machine Learning

Institute: BITS Pilani

Year: 2026

Supervisor: Dr. Dipti Jadhav

Abstract

Anaerobic digestion (AD) is a mature biological process used worldwide to convert organic waste into biogas, primarily composed of methane and carbon dioxide. The Anaerobic Digestion Model No. 1 (ADM1) provides a mechanistic framework to simulate biochemical reactions in AD systems; however, its computational complexity limits real-time deployment. Conversely, machine learning (ML) approaches enable accurate prediction of biogas production using operational data but often lack physical interpretability and consistency.

This dissertation proposes the **Green Fuel Intelligence System (GFIS)** — a physics-guided AI digital twin framework that integrates mechanistic modeling, machine learning, and physics-informed constraints. The system aims to predict methane yield, estimate hidden process variables, optimize operational conditions, and provide real-time decision support.

By combining ADM1-based synthetic data, hybrid AI models (LSTM and XGBoost), and a digital twin architecture, GFIS offers a scalable and intelligent solution for next-generation biogas systems. The proposed system will be implemented as a web-based platform over a 16-week development cycle.

Table of Contents

1. Introduction
2. Background and Literature Review
 - 2.1 Anaerobic Digestion
 - 2.2 ADM1 Model
 - 2.3 Machine Learning in Biogas Systems

2.4 Digital Twin Concept
2.5 Physics-Informed Machine Learning
3. Problem Statement
4. Objectives
5. Theoretical Foundations
6. Proposed Methodology
7. System Architecture
8. Implementation Plan
9. Expected Contributions
10. Conclusion
11. References

1. Introduction

The increasing demand for renewable energy and sustainable waste management has led to significant interest in biogas production through anaerobic digestion. Despite its widespread use, optimizing biogas systems remains a complex challenge due to nonlinear biological interactions and varying feedstock compositions.

Recent advancements in artificial intelligence and digital twin technologies provide new opportunities to enhance process efficiency, predict system behavior, and enable intelligent control mechanisms. This dissertation introduces GFIS, a unified framework combining AI, physics, and simulation to transform traditional biogas systems into intelligent, adaptive systems.

2. Background and Literature Review

2.1 Anaerobic Digestion

Anaerobic digestion is a biological process in which microorganisms break down organic matter in the absence of oxygen. The process consists of four stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis. These stages collectively convert complex organic materials into methane-rich biogas.

2.2 ADM1 Model

The Anaerobic Digestion Model No. 1 (ADM1) is a comprehensive mechanistic model that describes biochemical reactions and microbial dynamics in AD systems. It includes multiple state variables and kinetic equations, providing detailed system representation but posing challenges for real-time applications.

2.3 Machine Learning in Biogas Systems

Machine learning models, particularly neural networks, have demonstrated strong performance in predicting biogas production using SCADA data. Studies show that deep learning models outperform traditional regression techniques, especially when high-frequency data is available.

2.4 Digital Twin Concept

A digital twin is a virtual replica of a physical system that continuously updates using real-time data. It enables simulation, monitoring, and optimization without interfering with the physical system.

2.5 Physics-Informed Machine Learning

Physics-informed machine learning integrates physical laws into data-driven models. This improves model accuracy, reduces data requirements, and ensures predictions remain physically consistent.

3. Problem Statement

Methane production in anaerobic digestion is influenced by multiple nonlinear factors including operational parameters and feedstock characteristics. Traditional models are either too complex (ADM1) or lack physical consistency (ML models).

There is a need for a hybrid framework that combines:

- Mechanistic modeling
- Data-driven learning
- Real-time simulation

GFIS addresses this gap by integrating physics-guided AI with digital twin technology.

4. Objectives

1. Develop a hybrid data pipeline combining synthetic and real-world datasets.
 2. Design a hybrid AI model using LSTM and XGBoost.
 3. Integrate physics-informed constraints into ML models.
 4. Build a soft sensor for hidden variable estimation.
 5. Develop a digital twin for simulation and optimization.
 6. Create a web-based platform for deployment.
 7. Evaluate system performance using standard metrics.
-

5. Theoretical Foundations

5.1 Anaerobic Digestion Process

AD involves microbial breakdown of organic material into methane and carbon dioxide through multiple biochemical stages.

5.2 ADM1 Model

ADM1 simulates reaction kinetics and microbial interactions, serving as a foundation for synthetic data generation.

5.3 Machine Learning Models

LSTM networks capture temporal dependencies, while XGBoost handles structured tabular data efficiently. Ensemble learning improves robustness.

5.4 Digital Twin

Digital twins provide predictive and prescriptive analytics through real-time system representation.

5.5 Physics-Informed Learning

Physical constraints are embedded into ML models to enforce realistic predictions.

6. Proposed Methodology

6.1 Data Layer

- Synthetic data generation using ADM1
- Collection of SCADA datasets
- Data preprocessing and feature engineering

6.2 Hybrid AI Engine

- LSTM for temporal modeling
- XGBoost for tabular modeling
- Ensemble approach for prediction

6.3 Physics-Informed Constraint

Incorporation of constraints to ensure predictions do not exceed theoretical methane limits.

6.4 Soft Sensor Layer

Estimation of hidden variables such as VFA and alkalinity using ML models.

6.5 Digital Twin Layer

Real-time simulation, prediction, and optimization of biogas systems.

6.6 Optimization Module

Use of optimization algorithms to maximize methane yield while maintaining stability.

7. System Architecture

The GFIS system consists of four layers:

- 1. Data Layer
 - 2. Core AI Engine
 - 3. Soft Sensor Layer
 - 4. Digital Twin Layer
-

8. Implementation Plan (16 Weeks)

Week	Activities
1–2	Literature review and setup
3–4	Data collection and preprocessing
5–6	Model development
7–8	Physics integration
9–10	Soft sensor development
11–12	Digital twin prototype
13–14	Platform development
15	Testing and evaluation
16	Documentation and submission

9. Expected Contributions

- Hybrid physics-AI model
 - Soft sensor for stability metrics
 - Digital twin platform
 - Dataset and open-source framework
 - Research publication
-

10. Conclusion

GFIS presents a novel integration of AI, physics, and digital twin technology for biogas optimization. The system has the potential to significantly improve prediction accuracy, operational efficiency, and decision-making in anaerobic digestion systems.

11. References

1. ADM1 Documentation – WaterTAP
 2. Biogas Prediction using ML – PMC
 3. Physics-Informed ML – PNNL
 4. Digital Twin Concept – Encord
-